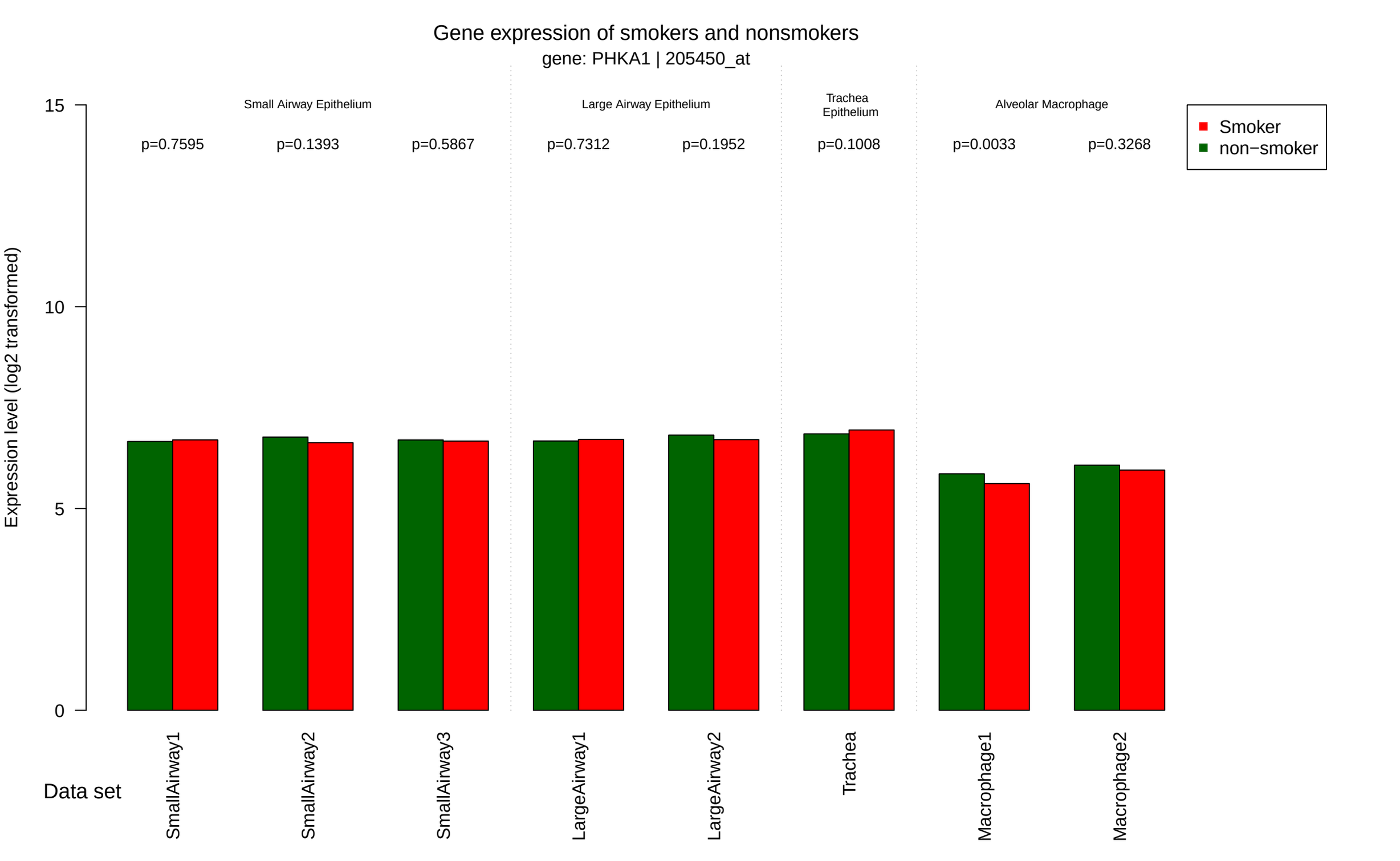
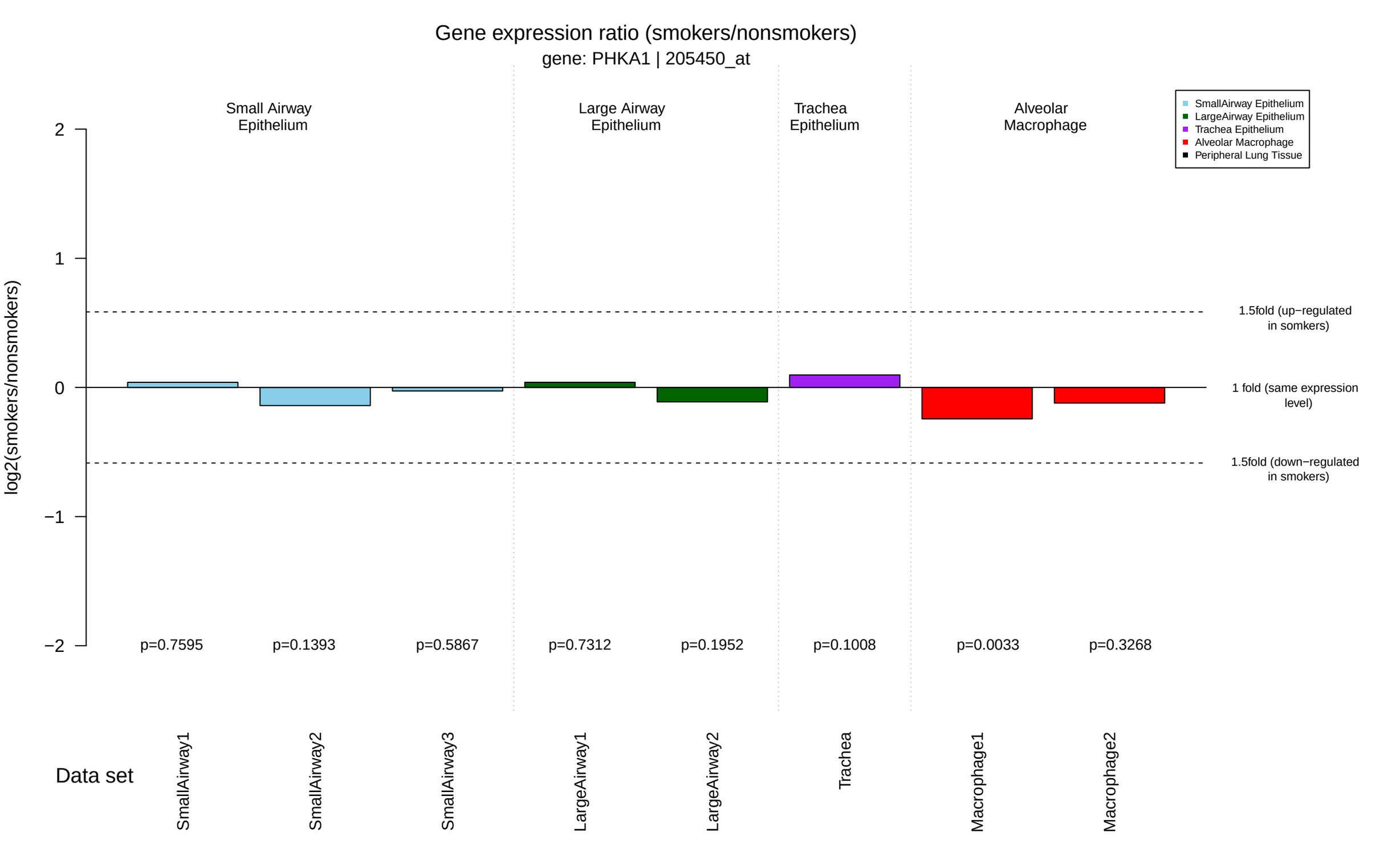
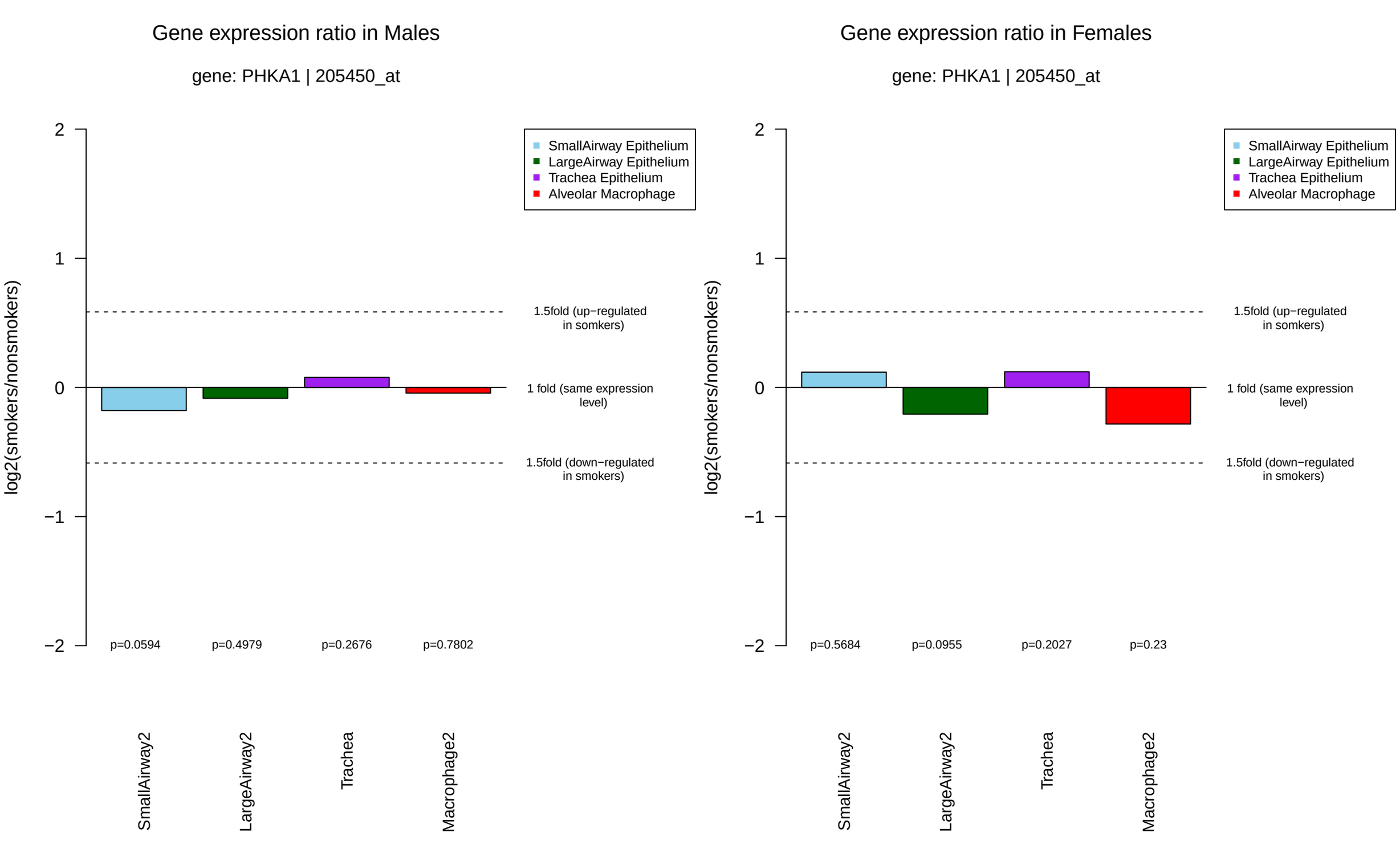
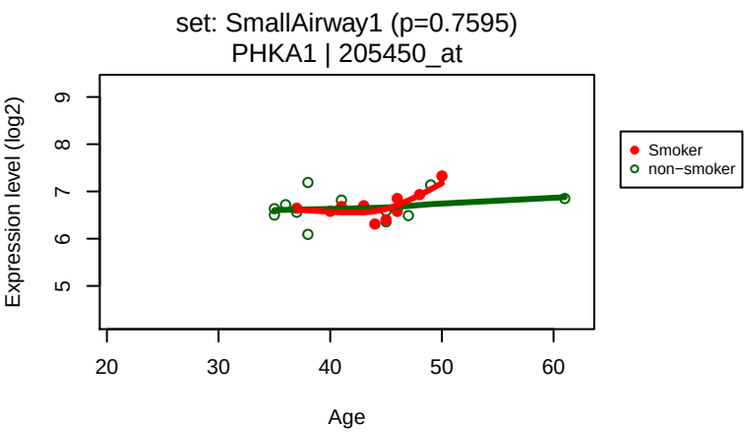
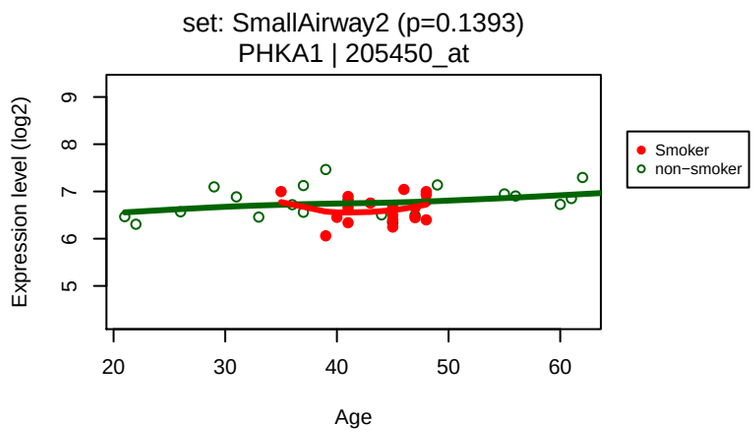
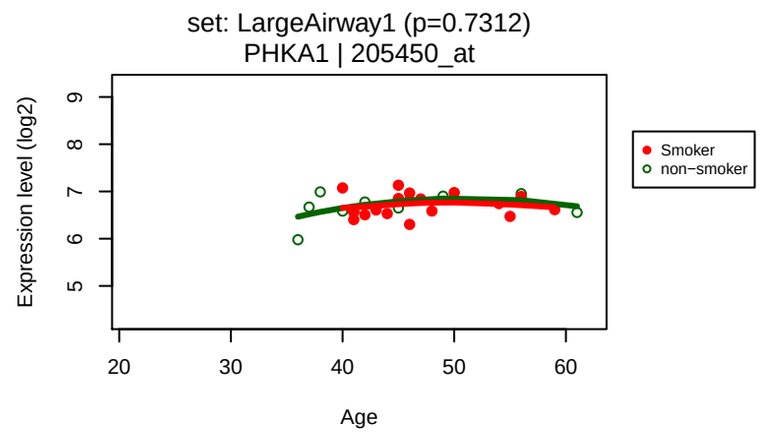

Gene expression of smokers and nonsmokers
gene: PHKA1 | 205450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7595
p=0.1393
p=0.5867
p=0.7312
p=0.1952
p=0.1008
p=0.0033
p=0.3268
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

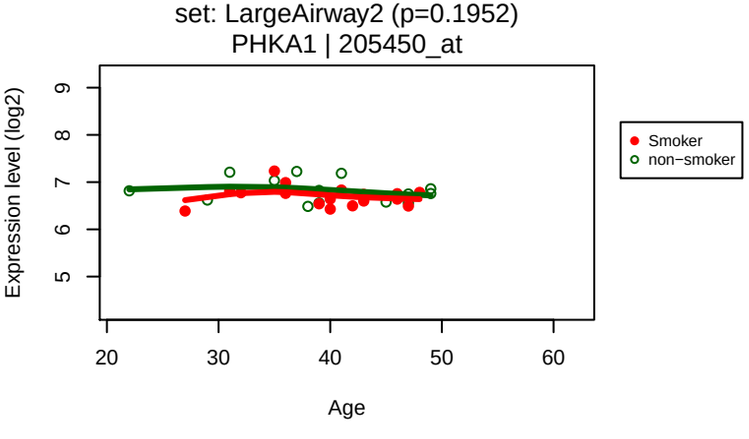

Gene expression ratio (smokers/nonsmokers)
gene: PHKA1 | 205450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7595
p=0.1393
p=0.5867
p=0.7312
p=0.1952
p=0.1008
p=0.0033
p=0.3268
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

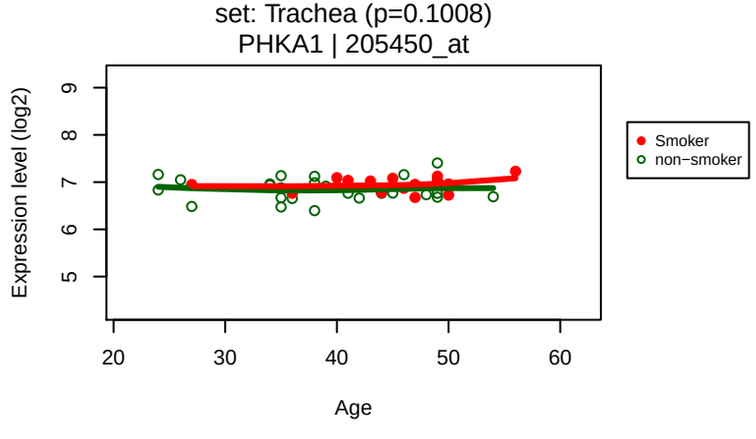

Gene expression ratio in Males
gene: PHKA1 | 205450_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0594
p=0.4979
p=0.2676
p=0.7802
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHKA1 | 205450_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5684
p=0.0955
p=0.2027
p=0.23
SmallAirway2
LargeAirway2
Trachea
Macrophage2

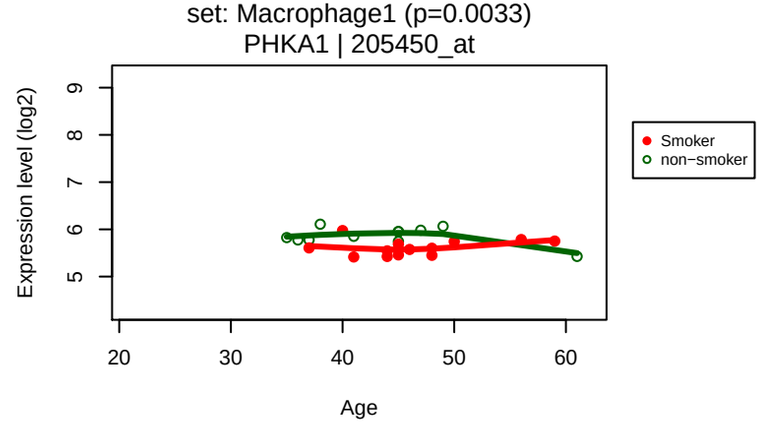

set: SmallAirway1 (p=0.7595)
PHKA1 | 205450_at
Expression level (log2)
Age
Smoker
non−smoker

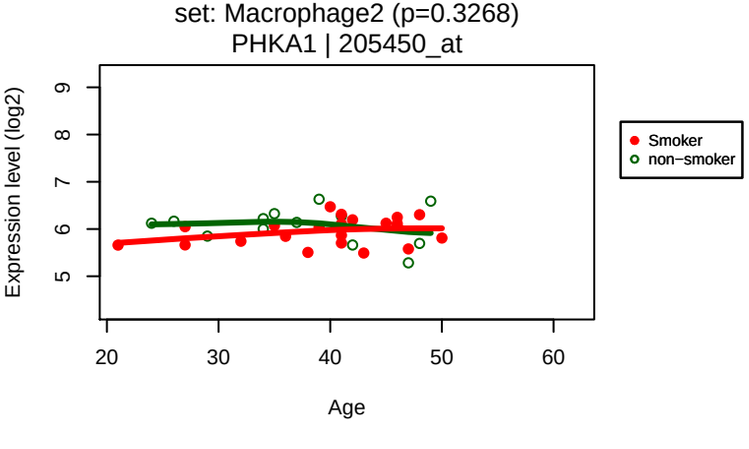

set: SmallAirway2 (p=0.1393)
PHKA1 | 205450_at
Expression level (log2)
Age
Smoker
non−smoker

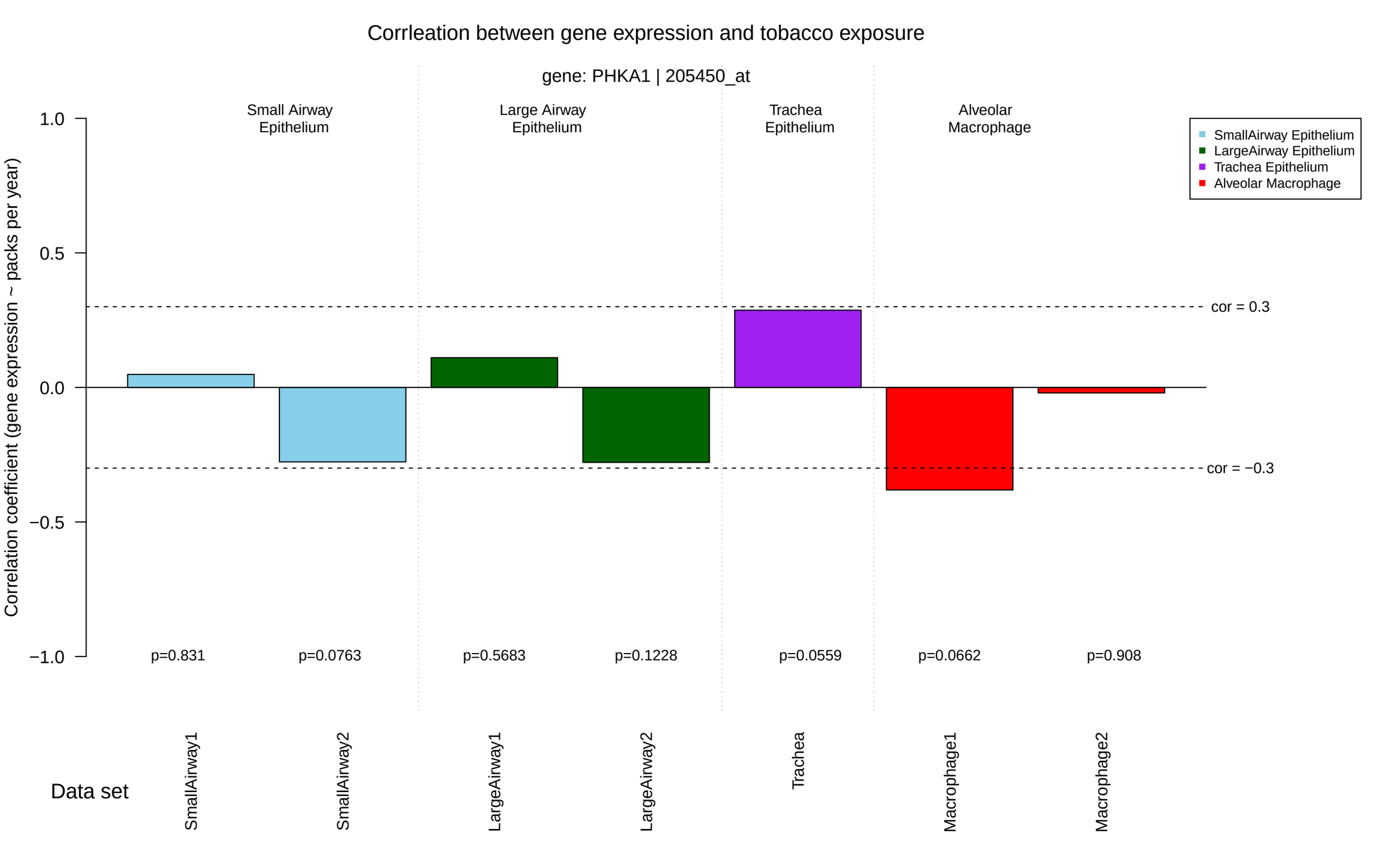

set: LargeAirway1 (p=0.7312)
PHKA1 | 205450_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1952)
PHKA1 | 205450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1008)
PHKA1 | 205450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0033)
PHKA1 | 205450_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3268)
PHKA1 | 205450_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHKA1 | 205450_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.831
p=0.0763
p=0.5683
p=0.1228
p=0.0559
p=0.0662
p=0.908
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2